

B.Tech IV Year I Semester (R20) Regular &amp; Supplementary Examinations December/January 2024/2025

**COST EFFECTIVE HOUSING TECHNIQUES**

(Common to EEE, ME, ECE, CSE, IT, FT, AI&amp;DS, CSE(AI), CSE(DS), CSE(AI&amp;ML), CSE(IoT), CSE(CS), CS&amp;D and AI&amp;ML)

Time: 3 hours

Max. Marks: 70

**PART – A**  
(Compulsory Question)

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- 1 Answer the following: (10 X 02 = 20 Marks)
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|-------------------------------------------------------------------------------------|----|
| (a) What is meant by residential density?                                           | 2M |
| (b) What are the various initiatives taken by government to address the urban poor? | 2M |
| (c) How GIS play major role in slum housing projects?                               | 2M |
| (d) Explain economical wall system in urban housing.                                | 2M |
| (e) Give a comments on low cost sanitation?                                         | 2M |
| (f) State the applications of Ferro cement?                                         | 2M |
| (g) What are the characteristics of Mud?                                            | 2M |
| (h) Explain any one rural housing programmes.                                       | 2M |
| (i) Specify the earthquake damages to houses.                                       | 2M |
| (j) What are the repairs of earthquake damaged non-engineered buildings?            | 2M |

**PART – B**

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the status of rural housing finance in India. 10M
- OR**
- 3 (a) Briefly explain the basic concepts and standards for housing programmes. 5M  
(b) What is the role of non-government organisations in slum housing? Explain. 5M
- 4 What is prefabrication? Explain few techniques that can be applied in roof construction. 10M
- OR**
- 5 Write a short notes on: 10M  
(i) Economic wall system,  
(ii) Fly ash gypsum thick for masonry.
- 6 (a) What is Ferro cement? How it could be applied for an alternate building material? 5M  
(b) How the industrial wastes are being used for alternate building material? Justify. 5M
- OR**
- 7 Explain in detail about the low-cost sanitation and domestic wall services in low cost infrastructure with their implementation standards as mentioned in IS codes. 10M
- 8 Illustrate the advantages of Mud housing technology with their impacts on plinth level during disaster and suggest for some improvement techniques in using Mud for construction. 10M
- OR**
- 9 (a) Explain fire treatment of thatch roof. 5M  
(b) Explain any two soil stabilisation techniques. 5M
- 10 (a) Suggest suitable measures for housing in disaster prone areas. 5M  
(b) What are the type of damages and repairs of non-engineered buildings? 5M
- OR**
- 11 Explain the repair and rehabilitation techniques used widely to resist the crack formation in the building during earthquake. 10M

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